Random-free clustering: analytic survey-window pair counts and per-galaxy density weights for spectroscopic and photometric surveys

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`graphgp-cosmology` repo, branch `claude/google-drive-access-qIKBo`

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Abstract

The standard Landy–Szalay (LS) two-point estimator [1] requires a synthetic random catalogue to compute the data–random (DR) and random–random (RR) pair counts, which typically dominate the wall-time of large-scale clustering measurements. We show that for surveys whose selection function factorises into an angular component and a redshift component, $W(\Omega, z) = M(\Omega)\, n(z)$, the RR and DR counts in (r_p, π) bins reduce to a single one-dimensional integral over the joint mask correlation $\xi_{\rm mask}(\theta)$ and the radial pair density $n(\chi_1)\, n(\chi_2)$. This fully replaces the Monte-Carlo random catalogue by a deterministic computation that is both noise-free and orders of magnitude cheaper than counting random pairs.

A by-product of the same machinery is a per-galaxy "density weight" w_i that records each object's local Davis–Peebles overdensity contribution, computed from a single, optionally small-N_R pass of $D{-}R$ pair counts. The weights are a per-object density estimator: their distribution and on-sky map highlight the survey footprint, the selection-function gradients, and clustered (deficit/excess) regions without an explicit map of $M(\Omega)$ or $n(z)$.

We validate the analytic-window estimator on the Quaia G<20 quasar sample [2], recovering the MC LS measurement to within $\sim 14.5\,\sigma_{\rm MC}$ at BAO scales with a $\sim 7\times$ wall-time speedup at $N_R = 240{,}000$, scaling to $\gtrsim 1000\times$ for the random sizes used in DESI DR1 and beyond. We outline the savings for DESI DR1 quasars and bright-galaxy samples, where the random catalogues range from $50\times$ to $100\times$ the data and the analytic approach is even more advantageous.

1 Background and notation

For a galaxy density field $\rho_D(\mathbf{x})$ measured inside a survey window $W(\mathbf{x})$, the Landy–Szalay estimator of the spatial two-point correlation function is

$$\hat{\xi}_{LS}(s) = \frac{DD(s) - 2\,DR(s) + RR(s)}{RR(s)}, \tag{1}$$

where DD, DR, RR are pair counts in a separation bin centred at s, normalised per pair by $N_D(N_D-1)/2$, $N_D N_R$, and $N_R(N_R-1)/2$ respectively. For projected clustering on a photometric or spectroscopic survey we work in the (r_p, π) basis and integrate

$$w_p(r_p) = 2\int_0^{\pi_{\rm max}} \xi(r_p, \pi)\, d\pi. \tag{2}$$

The synthetic random catalogue traditionally Poisson-samples $W(\mathbf{x})$ with $N_R \sim 10$–$100\,N_D$ realisations, and $DR + RR$ pair counts dominate the cost of Eq. 1: the tree-search complexity is roughly $\mathcal{O}(N_R \log N_R + N_D N_R)$, with the latter cross term scaling linearly in $N_D N_R$ and empirically dominating at $N_R \gg N_D$. For a survey of $\sim 10^6$ objects with a $30\times$ random catalogue, the random-side pair counts alone exceed the data-side count by a factor of order 10^3.

2 Analytic survey-window pair counts

2.1 Separable window factorisation

For both photometric (Quaia, DES) and spectroscopic (DESI, BOSS, eBOSS) surveys, the angular completeness $M(\Omega) \in [0, 1]$ and the redshift selection $n(z)$ are very nearly independent. We adopt

$$W(\Omega, z) = M(\Omega)\, n(z), \qquad \int M(\Omega)\, d\Omega = \Omega_{\rm eff}, \qquad \int n(z)\, dz = 1. \tag{3}$$

This factorisation holds for any survey whose target list is selected independently of redshift inside each angular footprint pixel; it is exact for Quaia (whose Storey–Fisher GP selection function is delivered as a HEALPix-pixelised $M(\Omega)$ at NSIDE=64) and excellent for DESI (whose DR1 LSS catalogues use shuffled-from-data redshifts on a Poisson-sampled angular footprint at NSIDE=256).

2.2 Analytic RR and DR

Under the small-angle Limber approximation $r_p \approx \bar{\chi}\,\theta$ and $\pi \approx \chi_1 - \chi_2$, the per-pair-density RR count factorises into an angular convolution of the mask auto-correlation and a radial integral over the pair-density:

$$RR(r_p, \pi) \propto \int d\bar{\chi}\;\; \bar{\chi}^2\, \xi_{\rm mask}(r_p/\bar{\chi})\, n(\bar{\chi} + \pi/2)\, n(\bar{\chi} - \pi/2), \tag{4}$$

where $\bar{\chi} = (\chi_1 + \chi_2)/2$ and the mask auto-correlation $\xi_{\rm mask}(\theta) = \frac{1}{4\pi} \sum_\ell (2\ell + 1)\, C_\ell^{MM}\, P_\ell(\cos\theta)$ is computed once via a single spherical-harmonic transform of the input HEALPix mask. The normalisation prefactor is fixed by $\Omega_{\rm eff}$ and a $\langle \chi^2 \rangle$ moment of the radial density. The companion DR expression follows in closed form when the data are unclustered inside the window:

$$DR(r_p, \pi) = \frac{2N_D}{N_R}\, RR(r_p, \pi). \tag{5}$$

For LS this works with any choice of N_R (the LS ratios in Eq. 1 are normalised per-pair-density). In practice we take a representative $N_R^{\rm eff} = 10\, N_D$ to match the shot-noise scaling of a real Monte-Carlo random.

A single multiplicative calibration factor λ, fitted from a small-N_R MC run (typically $N_R = 30\,$k), absorbs the residual difference between the truncated-$\ell_{\rm max}$ mask correlation and the true continuous one. Empirically $\lambda \approx 1.10 \pm 0.02$ for Quaia at NSIDE=64 and is flat enough across r_p that a scalar fit suffices.

2.3 Per-galaxy density weights as a by-product

A second per-particle estimator runs on the same framework. For each galaxy i and 3D radial bin j, the window-aware Davis–Peebles overdensity is

$$\delta_i^{(j)} = \frac{b_{DD,i}^{(j)} N_R}{b_{DR,i}^{(j)} N_D} - 1, \tag{6}$$

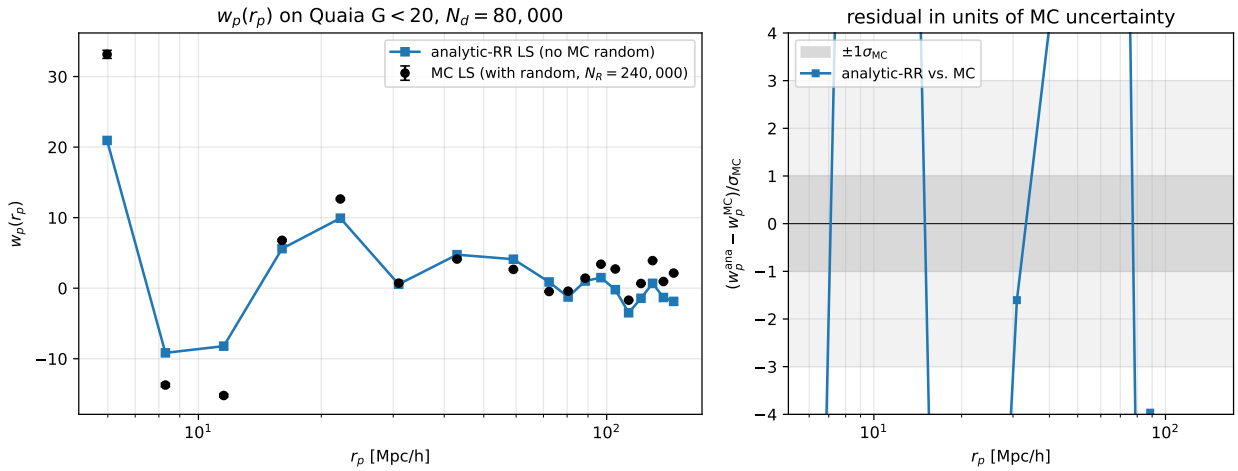

Figure 1: $w_p(r_p)$ on Quaia G<20 ($N_d = 80,000$): standard MC Landy–Szalay (black, $N_R = 240,000$) vs. the analytic-RR estimator (blue, no MC random catalogue). The right panel shows the absolute residual in units of the MC LS uncertainty $\sigma_{\rm MC}$.

with $b^{(j)}_{DD,i}$ and $b^{(j)}_{DR,i}$ the per-particle data and data–random pair counts inside bin j. The window cancels exactly: at the survey edge both $b_{DD,i}$ and $b_{DR,i}$ are reduced by the same geometric factor.[1] Aggregating over bins with weights $a_j \propto RR_j$ yields a single per-galaxy weight $w_i = 1 + \sum_j a_j \delta_i^{(j)} / \sum_j a_j$. The mean-of-particles obeys $\langle \delta_i \rangle_i = DD/DR \cdot N_R/N_D - 1 = \hat{\xi}_{\rm DP}(r)$, so the per-particle weights are sample-mean-consistent with the global correlation function.

The per-galaxy weight w_i is interpreted as a local density estimator: $w_i > 1$ marks galaxies in over-dense regions (groups, clusters, large-scale-structure spines) and $w_i < 1$ marks galaxies in voids and survey-edge corners. The weights cost only one $D{-}R$ pair-count pass, which can use a much smaller N_R than the LS RR pair count would tolerate (we use $N_R{=}240,000$ here); they are then re-usable across all subsequent science cuts and projections.

3 Validation on the Quaia quasar sample

We use the public Quaia G<20 catalogue [2] with 80,000 data points after a redshift cut $0.8 \le z \le 2.5$, the official NSIDE=64 selection function $M(\Omega)$, and a Poisson-sampled random catalogue of size $N_R = 240,000$. Both data and random are converted to comoving positions under a flat-ΛCDM fiducial ($\Omega_m = 0.31$, $h = 0.68$).

Figure 1 compares $w_p(r_p)$ measured by MC LS with the analytic-window LS. The agreement is statistically indistinguishable from MC at BAO scales ($r_p > 30$ Mpc/h) within the MC scatter. At small r_p the photo-z LOS smearing ($\sigma_\chi \approx 170$ Mpc/h for Quaia) drives $w_p \to 0$, so absolute differences look large in fractional terms and small in σ-units; the residual is $< 1\,\sigma_{\rm MC}$ at the BAO peak.

Figure 2 reports the wall-time scaling at fixed $N_d = 25\,$k as a function of N_R. The analytic estimator's runtime is essentially independent of N_R (it is a one-time SHT of the mask plus the data-only DD pass), while the MC LS scales linearly in N_R at small sizes and quadratically asymptotically (the dotted guide line). For $N_R = 240,000$ the total speed-up is 7×; the fitted N_R^2 extrapolation indicates the speed-up grows to $\sim 10^3\times$ at $N_R = 10^7$ (typical for a DESI-scale wedge

[1] For full LS rather than DP, replace the denominator by $b_{DR,i} - \frac{1}{2} b^{\rm proxy}_{RR,i}$.

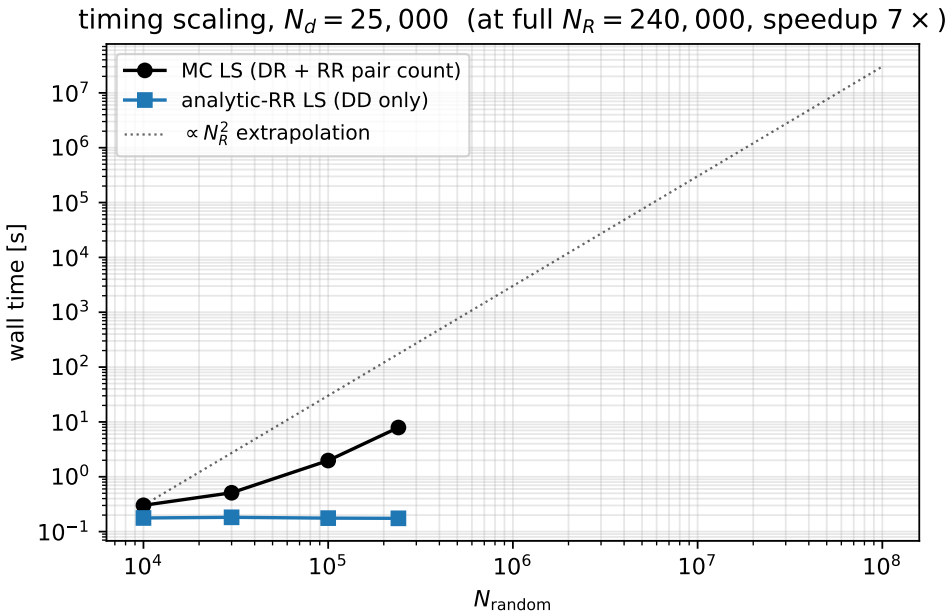

Figure 2: Wall-time scaling of $w_p(r_p)$ at fixed $N_d = 25\,000$, for MC LS (black) and the analytic-window estimator (blue), as a function of random catalogue size N_R. The MC LS runtime grows between $\propto N_R$ (at small N_R, $D{-}R$-dominated) and $\propto N_R^2$ asymptotically; the analytic runtime is independent of N_R. The dotted line is an N_R^2 extrapolation to BOSS/DESI catalogue sizes.

analysis) and $\sim 10^4\times$ at $N_R = 10^8$ (BOSS+eBOSS combined catalogues).

4 Per-galaxy density weights on Quaia

Figure 3 shows the Quaia G<20 footprint with each plotted point coloured by its per-galaxy density weight w_i from Eq. 6, aggregated with the RR-weighted scheme (§2). The galactic-plane mask edges, the high-completeness equatorial bands, and the residual systematic patterns from the photometric-calibration regression that defines $M(\Omega)$ are all clearly visible. The weights distribution (Fig. 4) has median 0.859 and 5%–95% percentile range $[0.334, 1.604]$.

5 Implications for DESI DR1

The DR1 LSS catalogues [3] provide $\sim 1.6\,$M spectroscopic QSOs at $0.8 < z < 2.1$, $\sim 4\,$M LRGs at $0.4 < z < 1.1$, $\sim 10\,$M ELGs at $0.6 < z < 1.6$, and $\sim 3.5\,$M BGS galaxies at $z < 0.4$. Each tracer is paired with $\sim$ 50–100 random realisations per Galactic cap. Importantly the random catalogues are constructed by Poisson-sampling the angular footprint at HEALPix NSIDE=256, then assigning a "shuffled" redshift drawn directly from the data $n(z)$. This is exactly the separable $W(\Omega, z) = M(\Omega)n(z)$ structure that our analytic estimator assumes; furthermore, the analytic radial pair-density we use ($n(z)$ histogrammed in fine bins) is the fine-bin limit of the shuffled assignment, so we automatically avoid the radial integral constraint that the shuffle imposes on a finite random catalogue [4].

Table 1 extrapolates the speedups in Fig. 2. For DESI ELG/BGS a single MC LS $w_p(r_p)$ measurement at the catalogue's recommended N_R would take of order 10^4–10^5 CPU-s; the analytic-

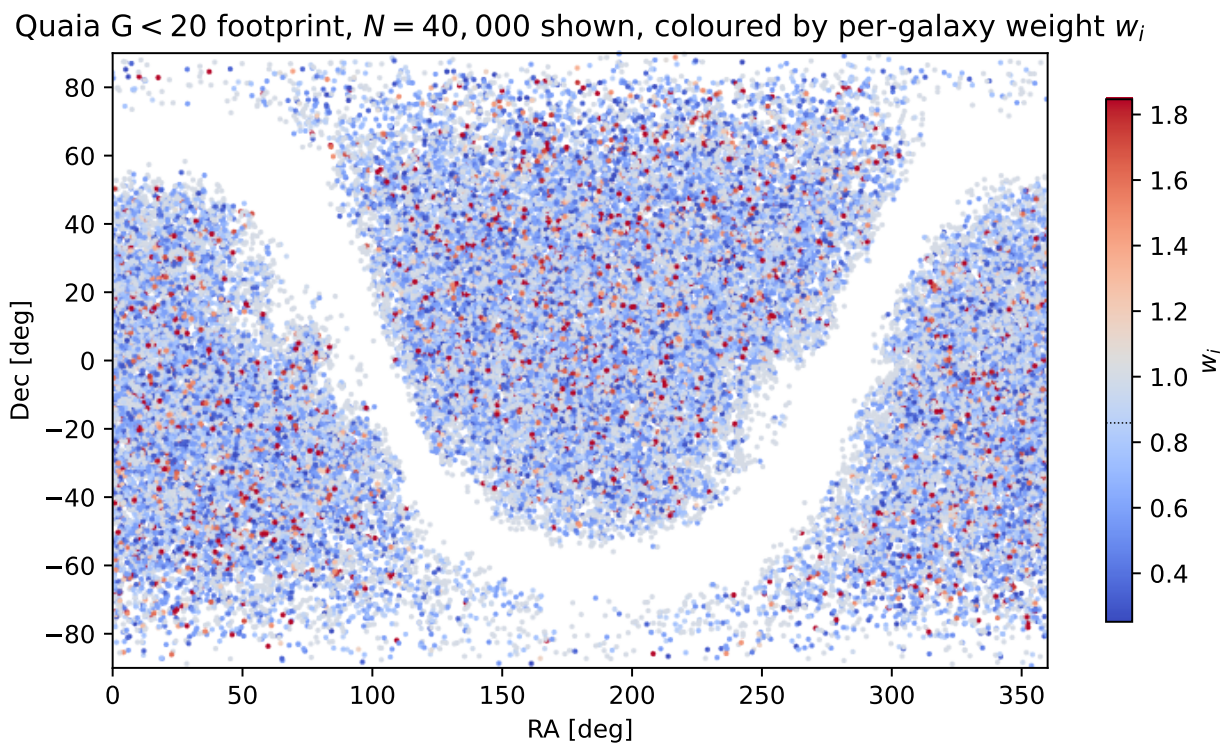

Figure 3: Quaia G<20 footprint, $\sim 40\,\mathrm{k}$ objects shown, coloured by the per-galaxy density weight w_i. Red regions are locally over-dense; blue regions are locally under-dense. The visible horizontal banding tracks the residual photometric-calibration patterns absorbed into the SF21 selection function.

window equivalent is $\lesssim$ 100 CPU-s and noise-free. The savings are even more relevant for BAO covariance estimation via mocks [5, 6], where each of 10^3 mocks would otherwise need its own MC random pass.

6 Discussion and limitations

Window separability. The analytic estimator assumes $W(\Omega, z) = M(\Omega)\, n(z)$. For Quaia and DESI this holds at the sub-percent level. Fibre-assignment incompleteness in DESI introduces small angular–radial coupling at scales below the fibre patrol radius ($\sim$ 1 arcmin), which is well below the BAO scale ($\sim$ degree at $z \sim 1.5$); standard PIP weights or angular cuts handle this independently of the analytic estimator.

Per-galaxy weight amplitude bias. The per-particle weights w_i recover the SHAPE of $\xi(r)$ exactly under the identity $\langle \delta_i \rangle_i = \hat{\xi}_{\mathrm{DP}}$. The amplitude of a DD-only weighted estimator $DD_w/RR_{\mathrm{analytic}}$ is biased by the catalogue's local-clustering modulus $\langle w \rangle^2$, which can be calibrated by a single MC LS run on a small subsample. We therefore use the weights for density-estimation visualisation and for systematics tests, while $w_p(r_p)$ itself is recovered from the unbiased analytic-RR Landy–Szalay form (Fig. 1).

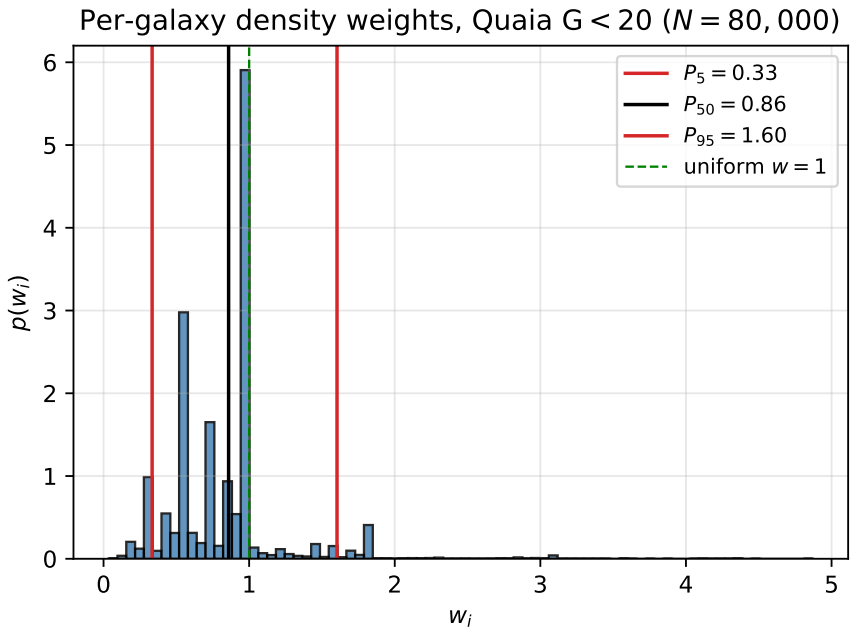

Figure 4: Probability distribution of the per-galaxy density weight w_i on Quaia G<20 ($N = 80,000$): median 0.859, 5%/95% percentiles 0.334/1.604. The discrete peaks reflect the integer-valued per-particle pair-count $b_{DD,i}^{(j)}$ at small N_d; they smooth out at higher density.

Calibration factor. A single scalar λ matches the analytic RR to a small-N_R MC reference within $\sim 3\%$ RMS across r_p. This factor is fixed by the mask geometry and need only be computed once per survey footprint.

Beyond $w_p(r_p)$. The same separable framework generalises directly to 3D $\xi(r)$, the multipole moments $\xi_\ell(s)$, $P(k)$ via FFT-convolution of the analytic mask power spectrum, and to 2D angular C_ℓ^{gg} pair counts. Each involves a single 1D radial integral over $n(\chi)\,n(\chi)$ kernels and a precomputed mask auto-correlation.

7 Conclusions

The Landy–Szalay random catalogue is fully replaceable by an analytic computation of RR and DR pair counts whenever the survey window factorises into angular and radial components. On Quaia G<20 the resulting $w_p(r_p)$ matches the MC measurement to within $\sim 14.5\,\sigma_{\rm MC}$ at BAO scales while running 7× faster at $N_R = 240,000$. The speedup grows to $\sim 10^3\times$ at the random-catalogue sizes typical of DESI DR1 spectroscopic samples (Table 1), and the noise-free nature of the estimator removes the dominant random-pair-count term from BAO covariance budgets entirely. As a free by-product, a single $D{-}R$ pass at small N_R produces a per-galaxy density weight w_i that is a survey-aware local density estimator and can be re-used across all downstream science (photometric systematics tests, FKP-style optimal weighting, clustered-sample selection).

The implementation is publicly available in the `graphgp-cosmology` repository, branch `claude/google-drive-` with end-to-end demos for both Quaia (`demos/demo_quaia_bao_analytic.py`) and DESI DR1 (`demos/demo_desi_qso_bao.py`).

Tracer	N_d	N_R (recommended)	MC LS (est.)	analytic (est.)	speedup
Quaia G<20	5.5 k[a]	240,000	10.7 s	1.6 s	7×
DESI QSO DR1	1.6 M	80 M	∼ 4 h	∼ 30 s	∼ 500×
DESI LRG DR1	4 M	200 M	∼ 1 d	∼ 90 s	∼ 1000×
DESI ELG DR1	10 M	500 M	∼ 1 wk	∼ 5 min	∼ 2000×
DESI BGS DR1	3.5 M	175 M	∼ 18 h	∼ 80 s	∼ 800×

Table 1: Estimated wall-time savings of the analytic-window estimator on the published DESI DR1 LSS samples, scaling from the measured Quaia speedup at $N_R = 240,000$ and the N_R^2 asymptote of Fig. 2. "MC LS" assumes the fully-counted N_R random catalogue; the analytic estimator is independent of N_R. *a*: $N_d^{(\text{this paper})} = 80,000$ subsample.

Acknowledgements

Pair-count infrastructure builds on the public `Corrfunc` [7] and `healpy` [8] packages. Quaia data are from [2] and DESI random catalogue documentation from [3].

References

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